Does a Mediterranean diet reduce primary cardiovascular events in adults at high cardiovascular risk?

An evidence review focused on the randomised trial base

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Short answer. Probably yes, for composite cardiovascular events and especially for stroke — but the randomised evidence in this specific population is **thinner and more fragile than the strength of the recommendation implies**. Essentially all of the direct randomised primary-prevention evidence comes from a single Spanish trial, PREDIMED, whose original report was retracted for randomisation failures and republished with revised estimates. That trial found fewer major cardiovascular events, no reduction in death from any cause, and a comparator that was itself an active dietary intervention. Independent syntheses of the same underlying trials reach materially different certainty verdicts. My assessment: **moderate confidence** that a Mediterranean dietary programme reduces composite cardiovascular events and stroke in high-risk adults; **low confidence** that it reduces cardiovascular or all-cause mortality in a primary-prevention setting; and **low confidence** that it is superior to other structured dietary programmes, particularly low-fat programmes.

1. The question, and what would count as an answer

The clinical claim under examination is narrow and should be kept narrow. It is not "is the Mediterranean diet healthy?" — an unremarkable proposition supported by a large and consistent observational literature. It is whether **assigning** adults who are at high cardiovascular risk but who have not yet had a cardiovascular event to a Mediterranean dietary pattern **causes** fewer subsequent hard cardiovascular events than assigning them to a comparator.

Answering that requires randomised trials that (i) enrol people without established cardiovascular disease, (ii) randomise to a Mediterranean dietary pattern, and (iii) count adjudicated clinical events — myocardial infarction, stroke, cardiovascular death — rather than lipids or blood pressure. That set of trials is remarkably small. The most recent Cochrane review of Mediterranean-style diets identified 30 randomised trials with 12,461 participants in total across primary and secondary prevention, but only a handful reported clinical events at all; the great majority contributed only risk-factor outcomes (Rees et al., 2019). For hard events in primary prevention, the field rests on one trial.

2. The trial that carries the question: PREDIMED

2.1 Design and headline result

PREDIMED was a multicentre Spanish trial that assigned 7,447 participants aged 55–80 years (57% women) who were at high cardiovascular risk but had no cardiovascular disease at enrolment to one of three arms: a Mediterranean diet supplemented with extra-virgin olive oil, a Mediterranean diet supplemented with mixed nuts, or a control diet consisting of advice to reduce dietary fat. Participants received quarterly educational sessions and, depending on allocation, free extra-virgin olive oil, free mixed nuts, or small non-food gifts. The primary endpoint was a composite of myocardial infarction, stroke, or death from cardiovascular causes. The trial was stopped after a median follow-up of 4.8 years on the basis of a prespecified interim analysis (Estruch et al., 2013, 2018).

A primary endpoint event occurred in 288 participants: 96 (3.8%) in the olive-oil arm, 83 (3.4%) in the nut arm, and 109 (4.4%) in the control arm. In the intention-to-treat analysis of the republished report, the hazard ratio was 0.69 (95% CI 0.53–0.91) for the olive-oil arm and 0.72 (95% CI 0.54–0.95) for the nut arm versus control (Estruch et al., 2018). Expressed in absolute terms from the reported percentages, that is a crude difference of roughly 0.6 and 1.0 percentage points over about five years — on the order of one event avoided for every 100 to 170 people offered the intervention for five years. The effect is real-looking but modest, and the trial counted only 288 events in total.

2.2 The retraction and republication

The 2013 report was retracted in 2018 and simultaneously republished. The investigators had identified protocol deviations: enrolment of household members without randomisation; assignment to a study group without randomisation for some participants at one of eleven study sites; and apparent inconsistent use of randomisation tables at another site. The republished analysis does not rely exclusively on the assumption that all participants were randomly assigned, and instead adjusts for baseline characteristics and propensity scores (Estruch et al., 2018; Retraction and Republication, 2018).

Two facts about this episode matter in opposite directions. First, the revised effect estimates barely moved: 0.70 → 0.69 and 0.72 → 0.72. Results were also similar after omitting the 1,588 participants whose allocations were known or suspected to have departed from protocol. That stability is genuinely reassuring about the arithmetic. Second, and less reassuringly, the repair changed the **nature** of the evidence: for a substantial fraction of the cohort the comparison is no longer protected by randomisation, and is instead protected by statistical adjustment. A propensity-adjusted comparison of non-randomly-allocated groups is an observational comparison. The trial's authority as a randomised experiment is therefore partially, not wholly, intact — a point pressed both in correspondence following republication (New England Journal of Medicine, 2018) and in independent commentary (Agarwal & Ioannidis, 2019).

2.3 What PREDIMED does not establish

Four features of the trial constrain what can be concluded from it, and they are separate from the integrity question.

The composite is driven by stroke. Cochrane's re-analysis of the PREDIMED comparison found a reduction in stroke (HR 0.60, 95% CI 0.45–0.80; a decrease from 24 to 14 per 1,000, graded moderate-quality evidence) but little or no effect on cardiovascular mortality (HR 0.81, 95% CI 0.50–1.32) or total mortality (HR 1.00, 95% CI 0.81–1.24), both graded low-quality (Rees et al., 2019). A composite endpoint whose signal sits almost entirely in one component supports a claim about that component, not about cardiovascular events generally.

The comparator was not "usual diet." The control arm received advice to reduce dietary fat — that is, an active dietary intervention that other randomised evidence indicates is itself beneficial. PREDIMED therefore estimates the incremental effect of a Mediterranean pattern over low-fat advice, not over doing nothing. In the trial's early years the control arm also received less intensive contact than the intervention arms, a difference the investigators have acknowledged and discussed (Guasch-Ferré et al., 2017). Some of the observed benefit may reflect intervention intensity rather than dietary composition.

The intervention was a package, not a diet. Participants received free extra-virgin olive oil or free nuts alongside repeated counselling. The trial cannot separate the effect of the dietary pattern from the effect of the supplied foods or of the behavioural support that delivered them.

Early stopping and a single setting. Trials halted early on an interim efficacy boundary tend to overestimate effect size. And the population was Spanish, aged 55–80, in a food environment where the intervention diet was culturally native. Transportability to populations for whom this pattern is a substantial dietary change is an assumption, not a finding.

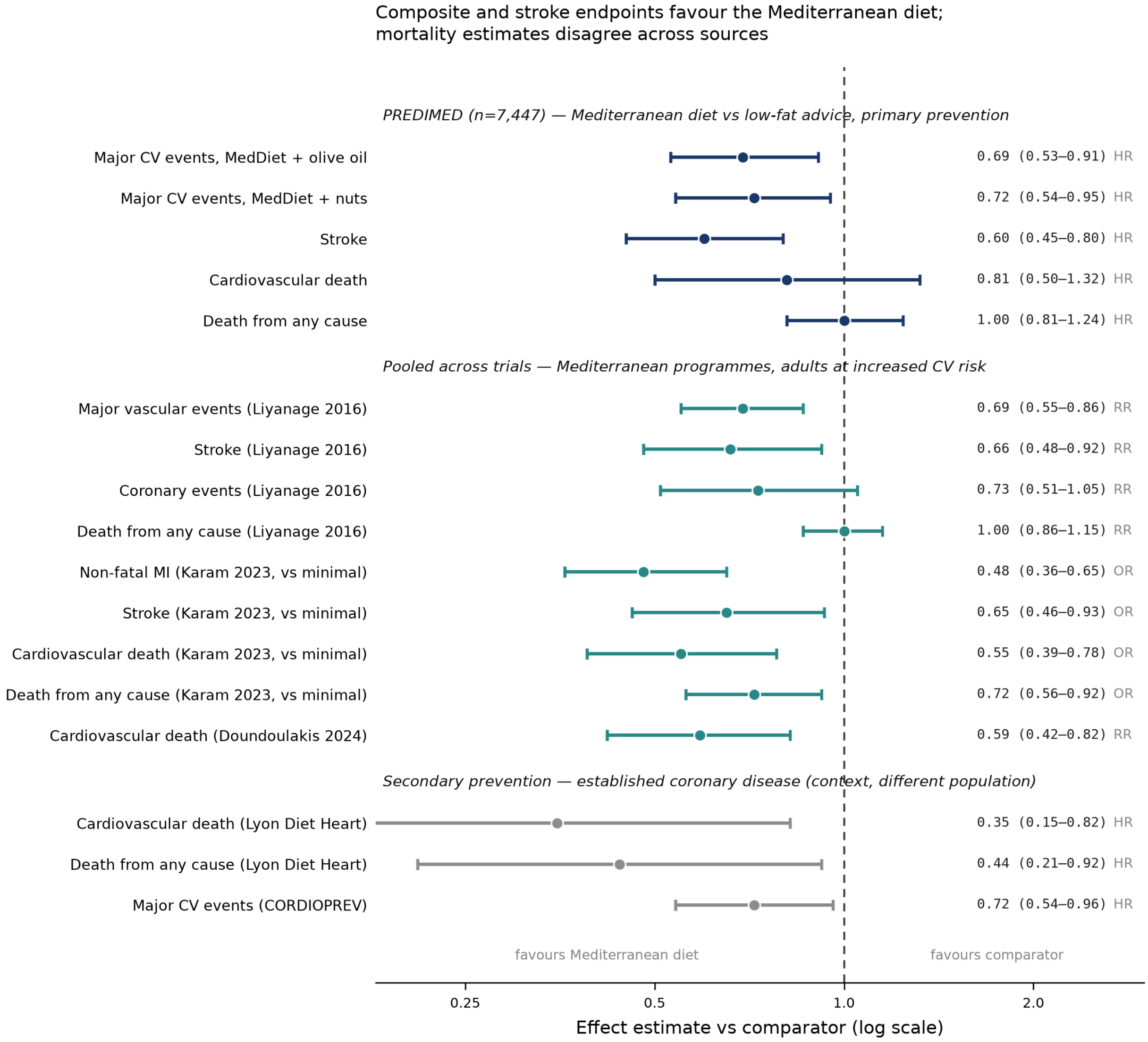

Figure 1: Effect estimates bearing on the question. Top block: the PREDIMED trial (Mediterranean diet versus low-fat advice, primary prevention in high-risk adults); composite-endpoint hazard ratios from the republished report (Estruch et al., 2018), component endpoints as re-analysed by Cochrane (Rees et al., 2019). Middle block: pooled and network estimates in adults at increased cardiovascular risk (Liyanage et al., 2016; Karam et al., 2023; Doundoulakis et al., 2024). Bottom block: secondary-prevention trials in patients with established coronary disease, shown for context only — a different population from the question at issue (Rees et al., 2019, for Lyon; Delgado-Lista et al., 2022, for CORDIOPREV). Effect measures differ across sources (HR, RR, OR) and are labelled per row; they are not interchangeable and the rows are not independent, since PREDIMED contributes to every pooled estimate shown. Liyanage estimates for vascular events, stroke and coronary events exclude the trial with data integrity concerns; the all-cause mortality estimate is for all trials combined.

3. Pooling the trials makes the picture less clear, not more

Three independent syntheses of largely the same trials reach conclusions that differ in strength, and the differences are instructive.

Cochrane (Rees et al., 2019) concluded that for primary prevention the evidence for benefit on clinical events was limited: moderate-quality for the stroke reduction, low-quality for mortality outcomes, and low or moderate quality for the modest risk-factor improvements. This is the most conservative of the three verdicts, and the most explicit that the primary-prevention clinical-event evidence is essentially one trial.

Liyanage et al. (2016) pooled six trials with 10,950 participants. Across all studies they found protection against major vascular events (RR 0.63, 95% CI 0.53–0.75), coronary events (0.65, 0.50–0.85), stroke (0.65, 0.48–0.88) and heart failure (0.30, 0.17–0.56), but not against all-cause mortality (1.00, 0.86–1.15) or cardiovascular mortality (0.90, 0.72–1.11). Critically, they flagged serious concerns about the integrity of the data in one large included trial (n = 1,000). **After excluding that trial, the benefit for vascular events (0.69, 0.55–0.86) and stroke (0.66, 0.48–0.92) persisted, but the apparent benefits for coronary events (0.73, 0.51–1.05) and heart failure (0.25, 0.05–1.17) disappeared.** Their own summary is appropriately hedged: the diet may protect against vascular disease, but the quantity and quality of evidence is limited and highly variable.

Karam et al. (2023), a BMJ network meta-analysis of 40 trials and 35,548 participants at increased cardiovascular risk across seven named dietary programmes (12 Mediterranean), reached the most favourable verdict. On moderate-certainty evidence, Mediterranean programmes were superior to minimal intervention for all-cause mortality (OR 0.72, 95% CI 0.56–0.92; 17 fewer per 1,000 over five years at intermediate risk), cardiovascular mortality (0.55, 0.39–0.78), stroke (0.65, 0.46–0.93) and non-fatal myocardial infarction (0.48, 0.36–0.65). A more recent network meta-analysis similarly found the Mediterranean diet to be the only dietary strategy reducing cardiovascular death (RR 0.59, 95% CI 0.42–0.82) among those compared (Doundoulakis et al., 2023).

Two caveats make the Karam result weaker support for the specific question than it first appears, and both are stated in the paper itself. First, the primary-prevention comparison of a Mediterranean diet against minimal intervention is **indirect**: it is constructed by combining PREDIMED (which compared Mediterranean with low-fat programmes) with primary-prevention trials comparing low-fat programmes against minimal intervention. There is no trial that directly randomises high-risk, event-free adults to a Mediterranean diet versus no dietary intervention. Second — and this cuts against the specific superiority of the Mediterranean pattern — the authors report **no convincing differences between Mediterranean and low-fat programmes** for mortality or non-fatal myocardial infarction. Since PREDIMED's comparator **was** a low-fat advice arm, the network's own head-to-head evidence is less enthusiastic than PREDIMED alone.

The methodological handling of data integrity also differs between reviews. Karam et al. performed sensitivity analyses excluding two studies because of concerns about data integrity, and where doing so substantially changed the findings they adopted the sensitivity analysis as the main result. Cochrane excluded two trials from its secondary-prevention analyses for published concerns about data reliability. Liyanage et al. showed explicitly how much turned on one such trial. Reviews that pool without this step will report larger and more confident effects than the underlying data support.

4. Findings that complicate or contradict the main conclusion

A large "Mediterranean" trial carries an expression of concern. The Indo-Mediterranean Diet Heart Study randomised 1,000 high-risk patients in India and reported a dramatic benefit: 39 versus 76 total cardiac endpoints (Singh et al., 2002). The Lancet subsequently issued an expression of concern about that study (Horton, 2005), against a documented background of difficulty in investigating suspected research fraud in this body of work (White, 2005). This is the trial whose removal dissolved the pooled coronary and heart-failure benefits in Liyanage et al. (2016). The point is not merely that one trial is doubtful; it is that a non-trivial share of the apparent consistency across meta-analyses has come from studies whose reliability has been formally questioned.

Mortality does not track the composite. In PREDIMED, all-cause mortality was flat (HR 1.00) and cardiovascular mortality did not reach significance. In Liyanage et al., all-cause mortality was flat across all trials (RR 1.00) and cardiovascular mortality was not significant. Karam et al. do find mortality benefits, but for Mediterranean-versus-minimal comparisons that are in part indirectly derived. If a diet genuinely prevented a meaningful share of major cardiovascular events in a high-risk population followed for five years, one would expect some mortality signal in the trial that observed those events. Its absence is a legitimate reason for caution, though five years and 288 events is also a thin basis on which to expect one.

Nutritional trial evidence has structural weaknesses that no single trial escapes. Dietary interventions cannot be blinded to participants; adherence is measured by self-report supplemented by biomarkers; "Mediterranean diet" is defined by adherence scores that vary between studies; and the field has attracted specific criticism for reproducibility and analytic practice (Ioannidis, 2018). These are not reasons to dismiss the evidence, but they place a ceiling on how certain any conclusion drawn from it can be, and they explain why two competent review teams can grade the same trials differently.

5. Adjacent evidence: supportive, but not a substitute

Secondary prevention. Two randomised trials in patients with established coronary disease support the dietary pattern. The Lyon Diet Heart Study (605 patients, 46 months) produced, on Cochrane's low-quality grading, adjusted reductions in cardiovascular mortality (HR 0.35, 95% CI 0.15–0.82) and total mortality (HR 0.44, 95% CI 0.21–0.92) (de Lorgeril et al., 1994, 1999; Rees et al., 2019). CORDIOPREV randomised 1,002 patients with coronary heart disease to a Mediterranean or a low-fat diet over seven years; the composite primary endpoint occurred in 87 versus 111 participants, with multivariable-adjusted hazard ratios across models ranging from 0.719 (95% CI 0.541–0.957) to 0.753 (0.568–0.998). The effect was evident in men but absent in the 175 women enrolled (Delgado-Lista et al., 2022). A 2025 synthesis in established cardiovascular disease reaches similar conclusions (Volpe et al., 2025). These trials strengthen biological plausibility, and CORDIOPREV is methodologically the cleanest Mediterranean-diet outcome trial available — but its population already has the disease that primary prevention aims to avoid, and its null result in women is itself a caution.

Observational cohorts. The cohort evidence is large, consistent, and directionally supportive: in the canonical Greek cohort of 22,043 adults, a two-point increment in a nine-point Mediterranean-diet score was associated with lower total mortality (HR 0.75, 95% CI 0.64–0.87) and lower coronary mortality (0.67, 0.47–0.94) (Trichopoulou et al., 2003), and recent reviews summarise a broadly concordant literature (Barbería-Latasa & Martínez-González, 2025). But adherence to a Mediterranean pattern is correlated with education, income, smoking, physical activity and healthcare access; effect sizes of this magnitude in nutritional cohorts are not reliably distinguishable from residual confounding. Cohorts cannot upgrade the randomised evidence — they can only fail to contradict it, which they do.

6. What would change the answer

PREDIMED-Plus is the successor trial: 6,874 participants with overweight or obesity and metabolic syndrome, randomised to an intensive lifestyle intervention of an energy-restricted Mediterranean diet plus physical activity and behavioural support, versus a control arm receiving **ad libitum** Mediterranean-diet advice, with hard cardiovascular endpoints as the primary outcome (Martínez-González et al., 2019). Its one-year results confirmed improvements in weight and cardiovascular risk factors (Salas-Salvadó et al., 2019). As of this review the primary endpoint has not been reported.

It is worth being clear about what that trial will and will not settle. Because **both** arms receive Mediterranean-diet advice, PREDIMED-Plus tests energy restriction, physical activity and behavioural intensity **on top of** a Mediterranean diet. It will not provide a second independent randomised test of the Mediterranean pattern itself against a non-Mediterranean comparator. The single-trial dependency identified in this review is therefore unlikely to be resolved by the trial most often cited as resolving it.

What would resolve it: a large, adequately powered, pre-registered primary-prevention trial in a non-Mediterranean food environment with independent endpoint adjudication and no free-food co-intervention. No such trial is currently reported.

7. Verdict, with explicit strength of evidence

Claim	Confidence	Basis and limiting factor
A Mediterranean dietary programme reduces composite major cardiovascular events in high-risk, event-free adults	Moderate	Direct randomised evidence from one trial (PREDIMED), consistent in pooled analyses; limited by single-trial dependency, partial loss of randomisation, early stopping, and an active comparator
It reduces stroke specifically	Moderate	The most robust component signal; graded moderate-quality by Cochrane and reproduced in two independent pooled analyses including after exclusion of the integrity-concern trial

Claim	Confidence	Basis and limiting factor
It reduces cardiovascular mortality	Low	Null in PREDIMED and in Liyanage et al.; significant in Karam et al. and Doundoulakis et al., partly on indirect comparisons. Sources disagree
It reduces all-cause mortality	Low	Null in the only direct primary-prevention trial (HR 1.00) and in one pooled analysis; positive in a network meta-analysis using indirect evidence
It is superior to other structured dietary programmes, notably low-fat	Low	Karam et al. report no convincing difference versus low-fat programmes for mortality or non-fatal MI; PREDIMED's own comparator was low-fat advice
The benefit generalises to non-Mediterranean populations	Low	No primary-prevention outcome trial outside a Mediterranean food setting; the one large non-European trial carries an expression of concern

Bottom line. The honest summary is that the recommendation is better supported than the alternative recommendations it competes with, and less well supported than its prominence in guidelines suggests. A Mediterranean dietary pattern is a reasonable, low-harm recommendation for adults at high cardiovascular risk, and the randomised evidence is more than merely absent — it points consistently in one direction for composite events and stroke. But it is one trial deep, that trial required retraction and partial de-randomisation, its mortality endpoints are null, and its comparator was itself an active diet. Anyone asserting that randomised trials have **established** that the Mediterranean diet prevents first cardiovascular events in high-risk adults is overstating what the trial base can carry. The claim that best fits the evidence is narrower: in high-risk adults, a Mediterranean-style dietary programme probably reduces major cardiovascular events, most clearly stroke, by an amount that is modest in absolute terms, and probably by no more than other structured dietary programmes achieve.

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